Dividing a proper fraction by a proper fraction

One rule now covers every case: multiply by the reciprocal of the divisor.

LESSON 27–29 | 3 × 40 MIN | MATEMATIKA 6, §7 | S7 7.4

LESSON CLOCK

20' | 30' | 30' | 30' | 10

The single rule	20 min
Cancelling	30 min
Mixed numbers	30 min
Bigger or smaller?	30 min
Homework	10 min

LEARNING OBJECTIVES

- Divide any fraction by any fraction.
- Divide mixed numbers by converting them first.
- Cancel before multiplying, and simplify the answer.
- Decide from the divisor whether the answer grows or shrinks.

TEXTBOOK

National	pp. 73–80
Cambridge	Stage 7, pp. 71–75
Workbook	Exercise 7.4

01 Explanation

The single rule

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c} = \frac{ad}{bc}$$

The three cases of the last two lessons were the same rule in disguise: a whole number n is $\frac{n}{1}$, whose reciprocal is $\frac{1}{n}$.

DIVISION	AS MULTIPLICATION	ANSWER
$\frac{1}{2} \div \frac{1}{4}$	$\frac{1}{2} \cdot \frac{4}{1}$	2
$\frac{3}{4} \div \frac{1}{2}$	$\frac{3}{4} \cdot \frac{2}{1}$	$\frac{3}{2} = 1\frac{1}{2}$
$\frac{2}{3} \div \frac{4}{5}$	$\frac{2}{3} \cdot \frac{5}{4}$	$\frac{5}{6}$
$\frac{5}{8} \div \frac{5}{8}$	$\frac{5}{8} \cdot \frac{8}{5}$	1

ANY NUMBER DIVIDED BY ITSELF IS 1

The last line is a check worth remembering: if the two fractions are equal, the answer must be 1, whatever the digits look like.

Cancelling

After turning the divisor over, cancel any numerator against any denominator before multiplying.

DIVISION	AFTER TURNING OVER	CANCEL	ANSWER
$\frac{3}{4} \div \frac{9}{8}$	$\frac{3}{4} \cdot \frac{8}{9}$	3	$\frac{2}{3}$
$\frac{5}{6} \div \frac{10}{3}$	$\frac{5}{6} \cdot \frac{3}{10}$	5	$\frac{1}{4}$
$\frac{7}{9} \div \frac{14}{3}$	$\frac{7}{9} \cdot \frac{3}{14}$	7	$\frac{1}{6}$

CANCEL ONLY AFTER THE DIVISOR HAS BEEN TURNED OVER

In $\frac{3}{4} \div \frac{9}{8}$ the 4 and the 8 cannot be cancelled while the ÷ sign is still there. Turn over first, then cancel — in that order, every time.

Mixed numbers

Convert every mixed number to an improper fraction **before** dividing. Nothing else in the method changes.

DIVISION	CONVERTED	ANSWER
$1\frac{1}{2} \div \frac{3}{4}$	$\frac{3}{2} \cdot \frac{4}{3}$	2
$2\frac{1}{4} \div 1\frac{1}{2}$	$\frac{9}{4} \cdot \frac{2}{3}$	$1\frac{1}{2}$
$3\frac{1}{3} \div \frac{5}{6}$	$\frac{10}{3} \cdot \frac{6}{5}$	4

A MIXED NUMBER CANNOT BE TURNED OVER AS IT STANDS

The reciprocal of $1\frac{1}{2}$ is $\frac{2}{3}$, not $1\frac{2}{1}$. Convert to $\frac{3}{2}$ first and the reciprocal is obvious.

Bigger or smaller?

DIVISOR	THE ANSWER, COMPARED WITH THE FIRST FRACTION	EXAMPLE
less than 1	larger	$\frac{3}{4} \div \frac{1}{2} = 1\frac{1}{2}$
equal to 1	unchanged	$\frac{3}{4} \div 1 = \frac{3}{4}$
greater than 1	smaller	$\frac{3}{4} \div \frac{3}{2} = \frac{1}{2}$

ONE GLANCE AT THE DIVISOR PREDICTS THE ANSWER

Comparing the divisor with 1 takes no calculation at all, and it catches the commonest error of the whole block — turning over the wrong fraction.

02 Worked examples

EXAMPLE 1 **Compute $\frac{2}{3} \div \frac{4}{5}$.**

1. The divisor is under 1, so expect an answer over $\frac{2}{3}$.

2. $\frac{2}{3} \cdot \frac{5}{4}$

   Turn over the divisor.

3. Cancel 2 against 4.

4. $= \frac{5}{6}$

ANSWER

$\frac{5}{6}$

EXAMPLE 2 **Compute $2\frac{1}{4} \div 1\frac{1}{2}$.**

1. Convert: $\frac{9}{4}$ and $\frac{3}{2}$.

2. $\frac{9}{4} \cdot \frac{2}{3}$

3. Cancel 9 with 3, and 2 with 4.

4. $= \frac{3}{2} = 1\frac{1}{2}$

ANSWER

$1\frac{1}{2}$

EXAMPLE 3 **Compute $\frac{5}{6} \div \frac{10}{3}$.**

1. $\frac{5}{6} \cdot \frac{3}{10}$

2. Cancel 5 with 10, and 3 with 6.

3. $= \frac{1}{4}$

   Smaller, because the divisor is over 1.

ANSWER

$\frac{1}{4}$

03 Interactive model

Put the four divisions of the block side by side — fraction by whole, whole by fraction, fraction by fraction — and ask what is common; the class states the single rule itself.

One rule for every division

$$\frac{2}{3} \div \frac{4}{5}$$

1. Divisor under

1 — the answer will be larger.

Estimate first.

2. $\frac{2}{3} \cdot \frac{5}{4}$

Turn over the divisor.

3. $\frac{5}{6}$

Cancel 2 against 4.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Division of fractions	Kasrlarni bo‘lish	Деление дробей
Reciprocal	Teskari son	Обратное число
Mixed number	Aralash son	Смешанное число
Improper fraction	Noto‘g‘ri kasr	Неправильная дробь
To cancel	Qisqartirish	Сокращать
Complex fraction	Qavatli kasr	Многоэтажная дробь
Estimate	Baholash	Оценка
Lowest terms	Qisqarmas ko‘rinish	Несократимый вид

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\frac{1}{2} \div \frac{1}{4}$ equals:

- $\frac{1}{8}$
- 2
- $\frac{1}{2}$
- 8

To divide by a fraction you:

- turn over the first
- turn over the divisor
- turn over both
- turn over neither

$\frac{2}{3} \div \frac{4}{5}$ equals:

- $\frac{8}{15}$
- $\frac{5}{6}$
- $\frac{6}{5}$
- $\frac{10}{12}$

A mixed number must first be:

- turned over
- converted to an improper fraction
- rounded
- doubled

$1\frac{1}{2} \div \frac{3}{4}$ equals:

- 2
- $\frac{9}{8}$
- $\frac{1}{2}$
- 1

A divisor greater than 1 makes the answer:

- larger
- smaller
- the same
- zero

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up — 7 PROBLEMS

1. $\frac{1}{2} \div \frac{1}{4}$

ANSWER 2

2. $\frac{3}{4} \div \frac{1}{2}$

ANSWER $1\frac{1}{2}$

3. $\frac{5}{8} \div \frac{5}{8}$

ANSWER 1

4. $\frac{2}{3} \div \frac{4}{5}$

ANSWER $\frac{5}{6}$

5. $\frac{1}{3} \div \frac{2}{3}$

ANSWER $\frac{1}{2}$

6. The reciprocal of $\frac{4}{9}$

ANSWER $\frac{9}{4}$

7. The reciprocal of $1\frac{1}{2}$

ANSWER $\frac{2}{3}$

Medium · the standard question

7 PROBLEMS

1. $\frac{3}{4} \div \frac{9}{8}$

ANSWER $\frac{2}{3}$

2. $\frac{5}{6} \div \frac{10}{3}$

ANSWER $\frac{1}{4}$

3. $\frac{7}{9} \div \frac{14}{3}$

ANSWER $\frac{1}{6}$

4. $1\frac{1}{2} \div \frac{3}{4}$

ANSWER 2

5. $2\frac{1}{4} \div 1\frac{1}{2}$

ANSWER $1\frac{1}{2}$

6. $3\frac{1}{3} \div \frac{5}{6}$

ANSWER 4

7. $\frac{3}{4} \div \frac{3}{2}$

ANSWER $\frac{1}{2}$

Hard · stretch and reason

7 PROBLEMS

1. $\frac{4}{5} \div \frac{2}{15}$

ANSWER 6

2. $2\frac{2}{3} \div 1\frac{1}{3}$

ANSWER 2

3. $\frac{9}{10} \div \frac{3}{5} \div \frac{3}{2}$

ANSWER 1

4. Without calculating: is $\frac{5}{7} \div \frac{8}{9}$ larger or smaller than $\frac{5}{7}$?

ANSWER Smaller

5. Without calculating: is $\frac{5}{7} \div \frac{2}{3}$ larger or smaller?

ANSWER Larger

6. A jug holds $\frac{3}{4}$ litre; how many $\frac{1}{8}$-litre cups fill it?

ANSWER 6

7. Why is $\frac{5}{8} \div \frac{5}{8} = 1$?

ANSWER Any number divided by itself is one

07 Homework

Homework — 5 tasks

Turn over the divisor first, then cancel — never the other way round.

1. Compute $\frac{3}{5} \div \frac{1}{2}$ and $\frac{4}{7} \div \frac{8}{9}$.

2. Compute $\frac{5}{6} \div \frac{5}{12}$.

3. Compute $1\frac{3}{4} \div \frac{7}{8}$.

4. Compute $2\frac{1}{2} \div 1\frac{1}{4}$.

5. Say, without calculating, whether $\frac{2}{3} \div \frac{5}{4}$ is larger or smaller than $\frac{2}{3}$, and why.